



**VISHWAKARMA INSTITUTE
OF TECHNOLOGY, PUNE**

MULTI-MODAL MIGRAINE RELIEF HEADSET SYSTEM

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PROBLEM STATEMENT

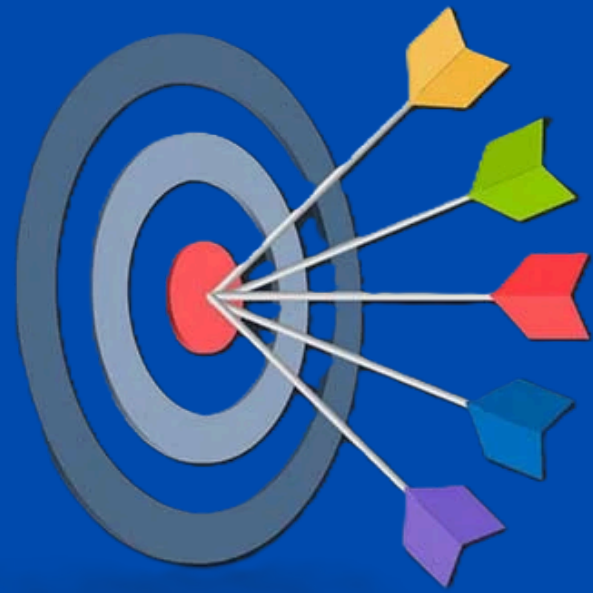
Migraines are commonly caused by stress, muscle tension, and sensory overload. Existing migraine-relief methods such as medication or single-therapy devices are often slow, inconvenient, or offer only partial relief. There is no affordable, portable, multi-modal solution that can provide real-time and personalized migraine relief.

Therefore, a system is needed that combines sound therapy, vibration therapy, and guided breathing into a single wearable device.

INTRODUCTION

- To provide natural, non-invasive, and immediate relief from migraine symptoms, we developed a Multi-Modal Migraine Relief Headset System. This system integrates sound therapy, vibration massage, and guided breathing exercises into a single smart wearable device.
- The headset includes built-in vibration motors attached to a forehead band for targeted muscle relaxation, while calming audio is delivered through the headphones to reduce stress and sensory overload. A structured 3-4-3 breathing cycle further helps stabilize the user's breathing and promotes relaxation.

OBJECTIVES:



- Develop a multi-modal wearable device that offers natural and non-invasive migraine relief.
- Integrate sound therapy, vibration massage, and guided breathing into one compact system.
- Provide a web-based control interface for easy customization and real-time adjustments.
- Deliver personalized therapy sessions based on user symptoms, stress level, and pain score.
- Offer an affordable, portable, and user-friendly solution for effective migraine management.

Overall Objective:

A unified, smart, and personalized system that delivers natural migraine relief through multi-modal therapy.

LITERATURE REVIEW

Title of Research Paper	Author(s)	Name of Journal/Conference	Date of Publication (Year)	Brief Summary of Work	Remarks & Gaps (Key findings, Technology, Tools, etc.)
1)A Proposed Model on Merging IoT Applications and Portable EEGs for Migraine Detection and Prevention	Akhila Jagarlapudi, Amey Patil, D. P. Rathod	IEEE International Conference on Distributed Computing, VLSI, Electrical Circuits and Robotics (DISCOVER)	2021	The paper proposes integrating IoT applications, portable EEG sensors, and VR headsets for effective migraine detection and prevention. It outlines EEG-based detection, Android app for trigger tracking, cloud & edge computing with RNN models, and VR for empathetic understanding.	Supports future scope (EEG integration). Gap: No therapy, no user interaction, no vibration/sound modality. Technologies used: Portable EEG, IoT, Android, Cloud & Edge Computing, TensorFlow, VR headsets, EEGLAB toolbox.
2) Predictive Analysis of Chronic Migraine Symptoms Detection and Management Using IoT and ML	S. Srinivasan, Dr. E. Punarselvam, Bharat Tidke, N. Gnanavel, Ahmed Shahebaaz, B. Meenakshi	International Conference on Advances in Modern Age Technologies for Health and Engineering Science (AMATHE 2025), IEEE	2025	Proposed an IoT-ML migraine monitoring system that collects user physiological/symptom data and predicts migraine trends.	Supports personalization part of our system. Gap: No physical therapy or multimodal relief offered—only monitoring and prediction.
3) Detection of Stress using Wearable Sensors in IoT Platform	Sreedevi Uday, C. Jyotsna, Amudha J.	Proceedings of the 2nd International Conference on Inventive Communication and Computational Technologies (ICICCT 2018), IEEE	2018	This research presents an IoT-based system for stress detection using wearable sensors. It uses a smart band (GSR sensor) and chest strap (ECG module) to monitor physiological signals like electrodermal activity and heart rate in real-time. Data is transmitted to ThingSpeak IoT platform for analysis and visualization using MATLAB.	Gap: No therapeutic intervention—only sensing. Significant correlation found between stress and physiological signals (GSR, HR). Technologies: IoT, wearable sensors. Tools: Arduino IDE, ESP8266 Wi-Fi module, ThingSpeak IoT

					platform, MATLAB analytics.
4) Design and Usability Evaluation of Wearable Therapeutic Devices for Migraine Relief: A Human-Centered Engineering Approach	Larry Campbell	Human-Centered Engineering	June 2025	This paper evaluates wearable therapeutic devices for migraine relief using a human-centered engineering approach. It emphasizes ergonomic fit, intuitive interfaces, comfort, usability, and user feedback. The study analyzed multiple devices using usability testing, expert heuristic evaluation, and patient feedback.	Key Findings: Human-centered design improves user adherence, satisfaction, and usability. Supports wearable-design principles of our headset. Gap: Did not combine multimodal therapy. Technologies: Wearable migraine devices (TENS, neuromodulation).
5) Music Medicine as a Component of Acute Migraine Attack Management	Pearl Angeli B. Diamante MD, Artemio A. Roxas Jr. MD, FPNA	Journal of Annals of Headache Medicine	2020	Demonstrates that music therapy reduces acute migraine intensity.	Supports our sound therapy module. Gap: Only audio therapy, no vibration or breathing.
6) Music Therapy for Stress Control and Autism	Divya Dand, Nithyashree V.C., Dr. Chayadevi M L	2019 1st International Conference on Advances in Information Technology (IEEE)	2019	The paper explores how music therapy, particularly Indian ragas, can help in reducing stress and assisting in autism treatment. It highlights experiments measuring changes in blood pressure and heart rate, the role of brain waves in stress detection, and the use of socially assistive robots to aid children with autism during therapy sessions.	Indian ragas positively affect mood, consciousness, and overall health. Supports sound therapy component of our system. Shows evidence that music reduces stress and autonomic arousal—key triggers for migraines. Gap: Focuses on autism, not migraine; does not explore multimodal therapeutic methods or IoT integration.

7) Effect of Local Vibration Therapy on Pain	Merve Yilmaz Menek ,Emre Dansuk,Umut Islam Tayboga	Journal of Clinical Medicine	2024	RCT showing vibration therapy reduces pain, stiffness, and muscle tension.	Strong evidence for vibration motor use. Gap: Not integrated into wearable IoT system.
8) Kinetic Oscillation Stimulation (KOS) for Preventive Treatment of Chronic Migraine	Jan Hoffmann , Holger Kaube, Florian Rimmele , Tim P. Jürgens, Markku Nissilä, Charly Gaul, Mikko Kallela, Petra Keski-Säntti, Marja-Liisa Sumelahti	Research Article Neurology	Feb 11, 2025	Demonstrates that vibration-based neuromodulation reduces migraine severity.	Supports vibration therapy justification. Gap: Medical-grade equipment, not portable or IoT-based.

METHODOLOGY

1.User Input Collection

Users enter pain intensity, stress level, symptoms, and pain location through the web application.

2.Therapy Session Generation

The system analyzes the inputs and automatically prepares a personalized therapy plan.

3.Guided Breathing Module

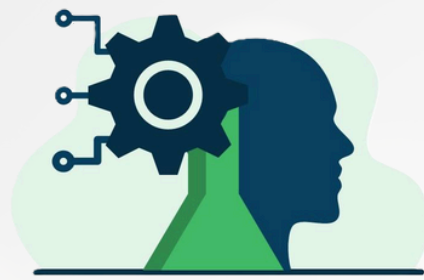
A 3-4-3 breathing exercise is presented to calm the user and reduce stress.

4.Sound & Vibration Control

Calming audio begins automatically, while vibration motors are controlled manually through the web interface.

5.Hardware Execution & Session Logging

The Arduino operates the vibration motors based on commands, and the system saves the session data for future reference.



TECHNOLOGY USED



HARDWARE TECHNOLOGIES :

- Arduino Uno – motor control, PWM intensity control
- Vibration Motors–forehead massage module
- Headphones–Sound therapy delivery
- Connecting Wires & Power Supply



SOFTWARE TECHNOLOGIES :

- HTML, CSS, JavaScript – frontend web interface
- Node.js / SQL and Php (backend) – Session handling
- Serial / HTTP Communication –Arduino Website control
- Audio Playback System – sound therapy module



SYSTEM ARCHITECTURE

➡ 1. Web Application Layer

The frontend collects user inputs, displays breathing guides, and controls sound & vibration settings, while the Node.js backend generates personalized therapy sessions.

➡ 2. Communication Layer

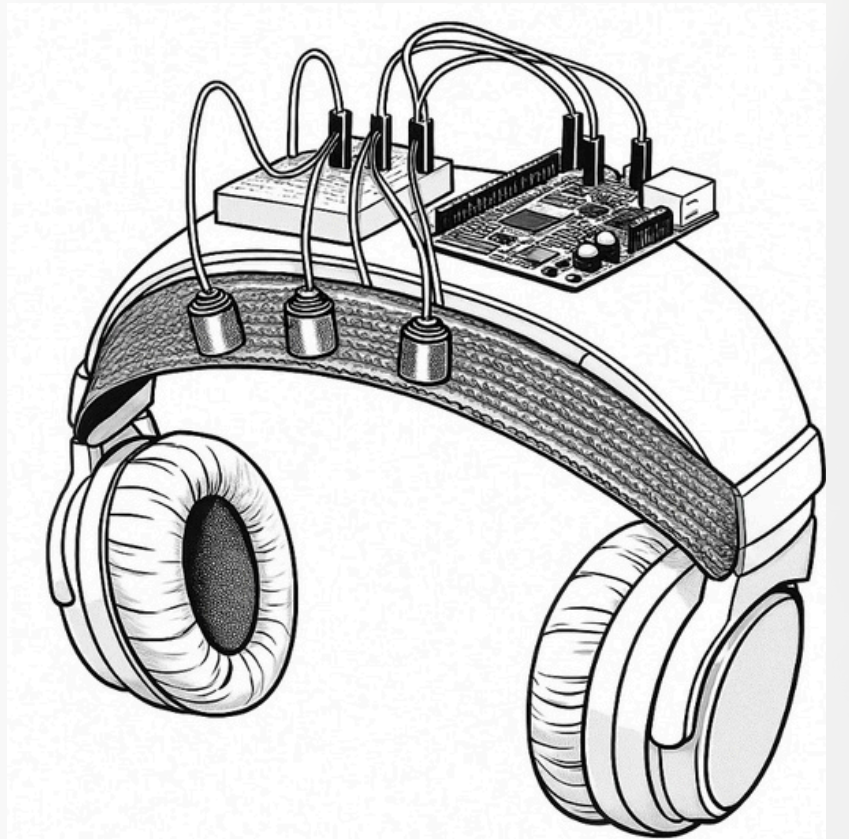
The Node.js server sends vibration commands to the Arduino Uno using Serial Communication through a USB A-to-B cable.

➡ 3. Device Control Layer

Arduino receives serial commands, processes vibration patterns, and drives the motors using PWM signals.

➡ 4. Hardware & Data Layer

Includes the vibration motors on the forehead band, headphones for sound therapy, and a database for storing session logs and user history.





FUTURE SCOPE & CONCLUSION

- AI Based Personalized Therapy –
Using machine learning to analyze user history and automatically suggest optimized therapy patterns.
- Advanced Sensors Integration –
Adding heart-rate, stress, or EEG sensors to detect migraine onset and trigger automatic therapy.
- Improved Hardware Design –
Creating a compact, lightweight PCB and custom-designed headband for comfort and commercial viability.

The migraine relief headset successfully combines sound therapy, vibration massage, and guided breathing into a simple, effective wearable device. With web-based control and Arduino hardware, it provides natural and personalized relief, showing strong potential for further development and practical applications.



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A top-down view of a desk with various items: a laptop in the upper center, a cup of coffee on the left, a pen and a small object in the center, a notebook on the right, a pair of glasses below the pen, and a large plant leaf in the bottom left corner. The background is a light, neutral color.

THANK YOU